GROUP ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

AICT009-4-2-IDA

INTRODUCTION TO DATA ANALYTICS

HAND OUT DATE: 22 MAY 2025

HAND IN DATE: 31 JULY 2025

WEIGHTAGE: 50%

INSTRUCTIONS TO CANDIDATES:

1. Submit your assignment at the administrative counter.
2. Students are advised to underpin their answers with the use of references (cited using the Harvard Name System of Referencing).
3. Late submissions will be awarded zero (0) unless Extenuating Circumstances (EC) are upheld.
4. Cases of plagiarism will be penalized.
5. The assignment should be bound in an appropriate style (comb bound or stapled).
6. Where the assignment should be submitted in both hardcopy and softcopy, the softcopy of the written assignment and source codes (where appropriate) should be on a CD in an envelope / CD cover and attached to the hardcopy.
7. You must obtain 50% overall to pass this module.

Title: Predictive Analytics for Diabetes Risk Using Health Indicators

Group Member	Student ID
1. KOO JUNN YUAN	TP082626
2. KUBERAN A/L KALAI SELVAN	TP082857
3. LOW WEI HAO	TP082257

Intake Code: UCDF2407ICT(DI)

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2. Business Goals & Objectives

2.1. Problem Statement:

Diabetes is among the most common non-communicable diseases affecting millions globally. This condition is chronic and viewed as irreversible. This project developed a predictive model to classify individuals as either diabetic or non-diabetic, using a range of medical and lifestyle indicators. Healthcare institutions may utilize this predictive tool to triage patients, screen patients, and intervene earlier to reduce the risk of diabetes.

2.2. Objectives:

- To clean and preprocess the Diabetes Health Indicators Dataset (Diabetes Health Indicator Dataset, 2022) by handling missing values, identifying outliers, and applying normalization where necessary.
- To explore the relationship between features and diabetes status using visual and statistical analysis.
- To develop and compare supervised classification models—including Logistic Regression, Random Forest, and Gradient Boosting—tailored to binary prediction.
- To evaluate the models using a confusion matrix, ROC-AUC score, precision, recall, and F1-score to determine the best-performing model.
- To identify and interpret influential features contributing to diabetes prediction, providing insights that can support clinical decision-making.

3. Domain Background

Diabetes is a metabolic disorder that can cause serious complications such as neuropathy, cardiovascular disease, retinopathy, and renal failure if not appropriately managed. The International Diabetes Federation (IDF) estimates that in 2021, 537 million adults were living with diabetes, with this expected to increase to 643 million by 2030 (Diabetes now affects one in 10 adults worldwide, 2021). The increasing incidence of diabetes, especially in low- and middle-income countries, requires cost-effective predictive approaches using existing health data. The early prediction and classification of diabetes risk allows for prevention before full clinical onset is established. Predictive analytics helps healthcare professionals identify individuals at high risk of developing diabetes and assist them in considering behaviour change and medical interventions.

4. Proposed Methodology

The proposed methodology for this project will follow the **CRISP-DM (Cross Industry Standard Process for Data Mining)** framework. Widely adopted across various industries, CRISP-DM offers a flexible yet structured approach to data analytics projects, supportive of iterative problem-solving, especially valuable for working with complex and inconsistent health data.

4.1. Justification for CRISP-DM

CRISP-DM allows for backtracking and iteration, which is relevant for healthcare datasets that include inconsistency or where deeper exploration may uncover additional matters. CRISP-DM supports documentation, reproducibility, and alignment with stakeholder expectations and engagement across the project lifecycle.

4.2. Dataset Description:

The dataset used in this analysis is the 'Diabetes Health Indicators Dataset' by Alex Teboul, published on Kaggle. It is based on data from the 2015 Behavioural Risk Factor Surveillance System (BRFSS), an annual telephone survey by the CDC for health-related information across the United States.

Source: (Diabetes Health Indicator Dataset, 2022)

Sample Size: 253,680 records

- **Target Variable:** `Diabetes_binary` with values 0 (No) and 1 (Diabetes)

- **Features:** 21 health-related indicators, including body mass index (BMI), smoking status, alcohol consumption, physical activity, stroke history, cholesterol check, sleep hours, and more.

4.3. Phases in CRISP-DM

1. **Business Understanding:**

   Define the core business problem and objectives: to develop a predictive model that classifies individuals as diabetic or non-diabetic, supporting early intervention in healthcare settings.

2. **Data Understanding:**

   Acquire and explore the dataset, assess data quality, and identify initial patterns or potential issues. This includes exploratory data analysis (EDA) to understand variable distributions and correlations.

3. **Data Preparation:**

   Clean the data, handle missing values or outliers, normalize numerical features (when needed), and perform feature engineering to prepare data for modelling.

4. **Modelling:**

   During modelling, class imbalance in the `Diabetes_binary` target variable will be assessed—initial analysis suggests that non-diabetic cases significantly outnumber diabetic ones. To address this, Synthetic Minority Oversampling Technique (SMOTE) will be applied to augment underrepresented instances, ensuring balanced model learning without introducing artificial bias. SMOTE was selected for its ability to preserve information while generating synthetic examples, as opposed to simple duplication or undersampling.

5. **Evaluation:**
   Assess how well the models are performing using performance metrics such as precision, recall, F1-score, and ROC-AUC. The focus will be on balancing accuracy with clinical relevance.
6. **Deployment (Hypothetically):**
   While this project does not include actual deployment, potential use cases—such as integrating the model into electronic health record (EHR) systems or patient screening tools—will be outlined. Additionally, ethical considerations such as data privacy, potential algorithmic bias, and the interpretability of model predictions will be acknowledged. These factors are crucial to ensure responsible AI deployment in clinical environments and to build trust among healthcare professionals and patients.

Figure 1: Phases in CRISP-DM

5. Tools & Technologies

5.1. Programming Language & Tools:

This project will be developed using **Python 3.x**, selected for its extensive ecosystem of data science libraries and strong community support. Key libraries include:

- **Pandas** and **NumPy** for data wrangling, transformation, and numerical computations.
- **Matplotlib** and **Seaborn** for generating visualizations to explore features, distributions, correlations, and class balance.
- **Scikit-learn** for building and evaluating supervised machine learning models.
- **Imbalanced-learn**, specifically the **SMOTE** techniques, for handling class imbalance in the target variable.
- **Google Colab** as the development platform, offering a cloud-based environment with built-in libraries and GPU acceleration.
- **Streamlit** to create an interactive application interface, enabling users to engage with the final model intuitively.

6. Expected Outcomes

By the end of this project, the following outcomes and value-added benefits are anticipated:

- Development of a robust classification model capable of predicting diabetes status with at least **80% predictive accuracy.**
- A ranked list of key medical and lifestyle features (e.g., BMI, physical activity, sleep duration) that contribute most to diabetes risk.
- **Business and public health impact**, including:
    - A *hypothetical* reduction of up to **30% in unnecessary clinical screenings**, assuming successful integration of the model in triage workflows.
    - Improved targeting of patient outreach and prevention programs for at-risk individuals.
    - Evidence-based recommendations for public health policies addressing modifiable risk factors.
- A **reusable and scalable analytics pipeline** that can be extended to newer datasets or potentially integrated into hospital information systems for clinical decision support.

7. References

Diabetes Health Indicator Dataset. (2022, March 10). Retrieved from Kaggle: https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset

Diabetes now affects one in 10 adults worldwide. (2021). Retrieved from International Diabetes Federation: https://idf.org/news/diabetes-now-affects-one-in-10-adults-worldwide/